Narration Script

LIGTA-S rationale video v1 · how MCS comes into play · version without narration

Length 2:35 · 1920 × 1080 · 30 fps · subtitles burned in · 294 words in four scenes, then a 10-second title card with no narration. Read each line over **LIGTA-S_Rationale_Video_v1_MCS_No_Narration.mp4**. Each line starts when its subtitle appears, and every visual cue is tied to a word, so match your pace to the subtitles.

How to record

1. Play the video and start speaking when each subtitle appears (**Start**).
2. Finish the line by **Finish by**. **Read time** is the time the line needs at an easy pace.
3. Record each scene as its own take. Scene windows: 0:00–0:21.6, 0:21.6–0:50.1, 0:50.1–1:15.1, 1:15.1–2:24.6, then the title card 2:24.6–2:35 (no narration).
4. The boom lands at 2:01.8. Read the conclusion (4.8 to 4.10) after it, with a full pause before the last line, then stay silent through the title card to 2:35.
5. Pace is about 2.3 words per second, with a short breath at each comma.

Say it this way

MCS: "M-C-S", letter by letter
LIGTA-S: "LIG-tas", like the Filipino word for safe
Mayon: mah-YON
Albay: al-BYE
Legazpi: leh-GAHS-pee
Camalig: kah-MAH-lig
lahar: LAH-har
Lola: LOH-lah
"test network", never "Albay", for the E1 numbers
RA 10821 and table values are shown on screen only. Do not read them.

Scene 1 · Family

0:00.0–0:21.6 · 43 words · bar: Safety

#	Start	Finish by	Read time	Narration (read this) · on screen
1.1	0:00.3	0:03.7	3.5 s	**January 2026. Mayon rises to Alert Level 3.** On screen: Mayon smokes on the right; a house and a family of four on the left; the badge "ALERT LEVEL 3" drops in under "January 2026".
1.2	0:04.2	0:09.8	5.7 s	**A family in the danger zone packs one bag and asks one question.** On screen: A packed bag appears beside the family; a dotted path leaves the house toward the lowlands.
1.3	0:10.3	0:15.0	4.8 s	**Where do we go, and will there be room for Lola?** On screen: Speech bubble "Where do we go?", then "Room for Lola?" over the grandmother with a cane.
1.4	0:15.6	0:20.2	4.8 s	**The nearest shelter is a school, and it is already full.** On screen: A school at the end of the path fills window by window; a red stamp "FULL" lands on it.

Scene 2 · Community

0:21.6–0:50.1 · 58 words · bar: Learning

#	Start	Finish by	Read time	Narration (read this) · on screen
2.1	0:21.9	0:24.4	2.6 s	**Zoom out to the whole province.** On screen: Albay as a field of 720 barangay dots in 18 towns; the danger zone glows around Mayon.
2.2	0:25.0	0:30.5	5.7 s	**In January, four schools turned into evacuation centers, and 3,424 learners were affected.** On screen: Card: "4 schools became evacuation centers · 3,424 learners affected" (counter), source line Philippine Daily Inquirer, 2026.
2.3	0:31.1	0:35.8	4.8 s	**By May, 34 classrooms in Camalig were sheltering about 1,315 people.** On screen: A grid of 34 classroom tiles fills; "34 classrooms · 488 families · about 1,315 people", source Daily Tribune, 2026.
2.4	0:36.3	0:44.4	8.3 s	**Sustainable Development Goal 4 asks for safe places to learn. Goal 11 asks us to protect people from disasters.** On screen: Chips slide in: "SDG 4.a · safe places to learn" and "SDG 11.5 · protect people from disasters".
2.5	0:45.0	0:48.8	3.9 s	**In Albay, both goals meet in the same classroom.** On screen: Both chips glide to the centre and meet over one classroom; RA 10821: "schools are a last resort" appears below.

Scene 3 · Stakes

0:50.1–1:15.1 · 51 words · bar: Time

#	Start	Finish by	Read time	Narration (read this) · on screen
3.1	0:50.4	0:54.3	3.9 s	**Then the alert rises, and minutes start to matter.** On screen: Badge changes to "ALERT LEVEL 4"; a clock counts minutes.
3.2	0:54.8	0:59.9	5.2 s	**Heavy rain sends lahar down the river channels and closes the bridges.** On screen: Two brown lahar channels cut across the road ring; bridges A1 to A4 and B1 to B3 turn red with a cross.
3.3	1:00.5	1:06.8	6.5 s	**Send everyone to the nearest site, and families arrive at shelters that are already full.** On screen: Arrows from every barangay to its nearest shelter; overflow counter "298 arrive at full shelters" (test network).
3.4	1:07.4	1:13.8	6.5 s	**So how do we shelter everyone, the vulnerable first, while keeping the most classrooms open?** On screen: Three lines build: "Everyone sheltered · Vulnerable first · Classrooms last" with a question mark.

Scene 4 · How MCS comes into play

1:15.1–2:24.6 · 142 words · bar: MCS

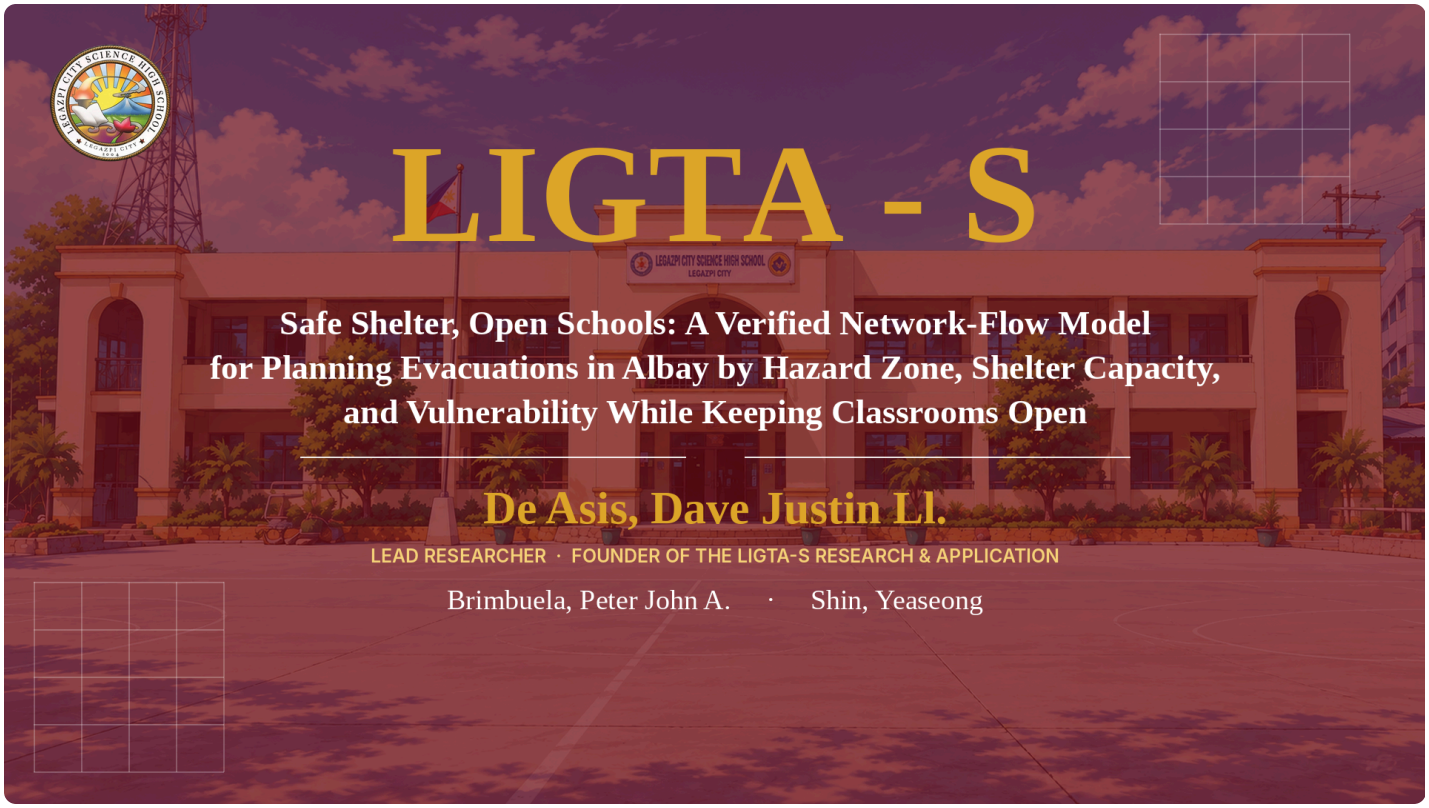
#	Start	Finish by	Read time	Narration (read this) · on screen

4.1	1:15.4	1:18.4	3.0 s	**This is where MCS comes into play.** On screen: The letters M, C and S drop in, then unfold into Mathematical, Computational, Sciences.
4.2	1:19.0	1:26.2	7.4 s	**Mathematics turns open data into a road graph, with travel times, capacities, and shelters in three tiers.** On screen: Chain lights "Open data → Graph"; panel MATHEMATICS: test network E1 draws in; G = (N, E), τ_e, u_e; tiers: non-school, school open space, classrooms (18 per room).
4.3	1:26.7	1:33.1	6.5 s	**The law's order becomes one lexicographic minimum-cost flow: vulnerable first, then everyone, then classrooms last.** On screen: Lexicographic objective and the penalties M_O and M_Y; evacuee flows animate from O1 to O3 into their shelters (Alert Level 3 plan).
4.4	1:33.7	1:40.1	6.5 s	**Computing solves it exactly in seconds, and a time-expanded network gives the exact clearance time.** On screen: Chain lights "Algorithm → Numbers"; panel COMPUTING: copies of the road network per minute; arrival curve F(T) climbs to 1,560; T* = 53 min, lower bound 34.
4.5	1:40.6	1:48.7	8.3 s	**Optimisation sites new Ligtas Pinoy Centers, and the greedy rule is guaranteed at least sixty-three percent of the best.** On screen: Chain lights "Optimisation"; panel OPTIMISATION: f(S_g) ≥ (1 − 1/e) f(S*) ≈ 0.632 f(S*); the south town lights inside the lahar pocket: 400 of 450 sheltered.
4.6	1:49.2	1:56.9	7.8 s	**On the test network, the plan closes fourteen classrooms instead of forty, for under two minutes more travel.** On screen: Counters: classrooms closed 40 → 14; learners displaced 1,702 → 532; extra mean trip +1.70 min. Label "test network E1 (synthetic)".
4.7	1:57.5	2:00.9	3.5 s	**Ten results, proven, then verified against independent code.** On screen: Chain lights "Proof"; ten proof squares turn into green checks; 300/300 plans, 660/660 networks, 4,796 checks, 0 violations.
4.8	2:02.0	2:05.0	3.0 s	**LIGTA-S will not build one more shelter.** On screen: Boom. Screen darkens; the line appears in white.
4.9	2:05.5	2:13.6	8.3 s	**But when the alert rises, it will tell every family where to go, and keep the most classrooms open.** On screen: The line appears in gold.
4.10	2:15.0	2:22.2	7.4 s	**[pause] Distance tells you where the shelter is. A plan tells you if there is room, in time.** On screen: Two lines: "Distance tells you where the shelter is." (white), then "A plan tells you if there is room, in time." (gold).

Title card · Researchers — 2:24.6–2:35 · no narration

#	Start	Finish by	Read time	Narration (read this) · on screen

5.1	2:24.6	2:35	—	**[No narration. Stay silent; let the music and the title card play.]** On screen: the school building under a deep red overlay, the school seal top-left and a white corner grid; the gold letters LIGTA - S drop in one by one with a hit, then a gold shine; the full paper title in white; two thin white lines draw outward; "De Asis, Dave Justin Ll." in large gold with "LEAD RESEARCHER · FOUNDER OF THE LIGTA-S RESEARCH & APPLICATION"; then "Brimbuela, Peter John A. · Shin, Yeaseong" in white. Fade to black at 2:35.



Title card, final frame (also supplied as LIGTA-S_Title_Page_1920×1080.png and LIGTA-S_Title_Page_3840×2160.png).

Clean read

The same narration without the production notes. Use it as a teleprompter or to rehearse. A slash marks a short breath.

SCENE 1 · FAMILY · 0:00.0–0:21.6

0:00.3 January 2026. Mayon rises to Alert Level 3.

0:04.2 A family in the danger zone packs one bag and asks one question.

0:10.3 Where do we go, / and will there be room for Lola?

0:15.6 The nearest shelter is a school, / and it is already full.

SCENE 2 · COMMUNITY · 0:21.6–0:50.1

0:21.9 Zoom out to the whole province.

0:25.0 In January, / four schools turned into evacuation centers, / and 3,424 learners were affected.

0:31.1 By May, / 34 classrooms in Camalig were sheltering about 1,315 people.

0:36.3 Sustainable Development Goal 4 asks for safe places to learn. Goal 11 asks us to protect people from disasters.

0:45.0 In Albay, / both goals meet in the same classroom.

SCENE 3 · STAKES · 0:50.1–1:15.1

0:50.4 Then the alert rises, / and minutes start to matter.

0:54.8 Heavy rain sends lahar down the river channels and closes the bridges.

1:00.5 Send everyone to the nearest site, / and families arrive at shelters that are already full.

1:07.4 So how do we shelter everyone, / the vulnerable first, / while keeping the most classrooms open?

SCENE 4 · HOW MCS COMES INTO PLAY · 1:15.1–2:24.6

1:15.4 This is where MCS comes into play.

1:19.0 Mathematics turns open data into a road graph, / with travel times, / capacities, / and shelters in three tiers.

1:26.7 The law's order becomes one lexicographic minimum-cost flow: vulnerable first, / then everyone, / then classrooms last.

1:33.7 Computing solves it exactly in seconds, / and a time-expanded network gives the exact clearance time.

1:40.6 Optimisation sites new Ligtas Pinoy Centers, / and the greedy rule is guaranteed at least sixty-three percent of the best.

1:49.2 On the test network, / the plan closes fourteen classrooms instead of forty, / for under two minutes more travel.

1:57.5 Ten results, / proven, / then verified against independent code.

2:02.0 LIGTA-S will not build one more shelter.

2:05.5 But when the alert rises, / it will tell every family where to go, / and keep the most classrooms open.

2:15.0 [pause] Distance tells you where the shelter is. A plan tells you if there is room, / in time.

TITLE CARD · RESEARCHERS · 2:24.6–2:35

[Silent. Title card: LIGTA-S · De Asis, Dave Justin Ll., Lead Researcher · Founder · Brimbuela, Peter John A. · Shin, Yeaseong. Ends 2:35.]